2(a)	No resultant external force acting on the two colliding / interacting objects.	B1
(b)(i)	Force on B = rate of change of momentum of B $= \frac{(24-20)\times 10^3}{3.0-1.5}$ = 2667 ≈ 2700 N	M1 A1
(b)(ii)	Gradient is equal to the rate of change of momentum of the truck and this is also equal to the impact force exerted on the trucks according to Newton's 2nd law Since the impact forces between the two trucks are an action-reaction pair and and are of equal magnitude but acts in opposite directions, the gradients have the same magnitude but opposite signs.	B1 B1
(b)(iii)	Impulse acting on each truck (during the collision) is equal to the change in momentum of the truck. Since the impulse on the two trucks are equal and opposite, it follows that the gain in momentum of one truck must be equal to the lost in momentum of the other truck. Hence total momentum is conserved.	B1 B1
(b)(iv)	Total kinetic energy of the two trucks before collision $= \frac{(20{,}000)^2}{2\times4000} + \frac{(16{,}000)^2}{2\times2000} = 114$ kJ Total kinetic energy after collision $= \frac{(24{,}000)^2}{2\times4000} + \frac{(12{,}000)^2}{2\times2000} = 108$ kJ [C1- for calculation of KE before and after collision] Change in kinetic energy = - 6000 J (Perfectly) Inelastic collision	 A1 B1